

Seif Sheikhelarab

Backend Engineer

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Profile

Backend engineer targeting junior/entry-level roles. Shipped 6+ production-grade systems spanning REST APIs, job queues, and data pipelines in TypeScript, Node.js, and PostgreSQL. Recent B.Sc. graduate in Communications & Computer Engineering (GPA 3.84/4).

Skills

Languages — TypeScript, JavaScript, Go, Python
Frameworks & Libraries — React, Express.js, Hono, Tailwind CSS
Data & Infrastructure — PostgreSQL, MongoDB, Redis, Kafka, Docker, Linux
Practices — REST APIs, WebSockets, RBAC, CI/CD, Observability, Zod

Experience

Freelance Client Project — Kiosk POS and Loyalty Platform Dec. 2025 – Apr. 2026

Node.js, TypeScript, Express.js, Prisma ORM

- Built a real-time transaction engine handling deposits, withdrawals, commissions, and transfers, with shadow wallets for unregistered users across 112 secured endpoints.
- Designed a multi-role auth system (Admin/Owner/Worker/Customer) with OTP passcodes and JWT session rotation.
- Engineered an admin dashboard with financial reporting, audit logging, KPI graphs, and suspicious-activity alerts across 13 feature modules and 17 data models.

Projects

Briefly — E-Commerce CRM System with AI Features (Graduation Project) Oct. 2025 – Jun. 2026

React.js, TypeScript, Bun, Express.js, PostgreSQL, Redis

Live Demo | GitHub | API Docs

- Ran async job processing with BullMQ across 8+ queues for CSV/XLSX import-export, RFM scoring, Shopify sync, and Meta omnichannel messaging, with Redis-backed persistence and retry handling.
- Implemented Full CRM functionality with subscription billing with Paymob, a Shopify integration with HMAC-verified webhooks, and CI/CD covering 400+ unit and integration tests on every push.

Catalyst — Real-Time Analytics Pipeline

Jun. 2026

TypeScript, Bun, Hono, Kafka, Redis, ClickHouse, TimescaleDB

GitHub

- Built 6 microservices on Kafka with CQRS-separated writes and reads, and dual API key/JWT authentication.
- Added production resilience with retry backoff, dead-letter queues, circuit breakers, Kafka offset management, and idempotency keys, plus observability via Prometheus, Grafana, Jaeger, and Loki.

latch-store — npm idempotency middleware

Jul. 2026

TypeScript, Express.js, Hono, Redis

npm | GitHub | Docs

- Published an npm middleware library that makes any mutating endpoint idempotent in one line, with memory and Redis store adapters, safe handling of concurrent duplicate requests, and generated API docs.

Education

Bachelor's Degree in Communications and Computer Engineering

Oct. 2021 – Jun. 2026

Faculty of Engineering - Mansoura University, GPA: 3.84/4 — Excellent degree with Honors

Extracurricular Activity

Backend Committee Trainee, CAT Reloaded (university student club; 2023 – 2025) | **Competitive Problem Solver, ICPC Mansoura** (2023 – 2024)

Certificates & Languages

AWS Cloud Foundations, AWS Academy | Cloud Computing Fundamentals, IBM SkillsBuild

Arabic (Native) | English (Fluent, C1) | French (Basic, A1)